

Research interests

Extremal combinatorics and graph theory, algebraic and probabilistic methods, discrete geometry, combinatorial number theory.

Employment

- **Research fellow**, Delft University of Technology (TU Delft) 09/2026–now
 - Mentor: Anurag Bishnoi
- **Research fellow**, Institute for Basic Science (ECOPRO) 01/2024–08/2026
 - Mentor: Hong Liu
- **Research fellow**, Alfréd Rényi Institute of Mathematics 09/2023–12/2023
 - Mentor: János Pach

Education

- **PhD in Mathematics**, Carnegie Mellon University 09/2018–05/2023
 - Advisor: Boris Bukh
 - Thesis: “Several problems in extremal combinatorics”
- **BS in Mathematics**, Peking University 09/2014–07/2018
 - Advisor: Chunwei Song
 - Thesis: “Three proofs of Roth’s theorem”
- **Middle and High School**, Beijing National Day School 09/2008–07/2014

Teaching experience

- **Teaching assistant at CMU** 09/2018–05/2023
 - 21-325 Probability, grading Spring 2023
 - 21-228 Discrete Mathematics, grading and recitations Spring 2022
 - 21-301 Combinatorics, grading Fall 2021

- 21-623 Complex Analysis (graduate), grading Spring 2021
- 21-295 Putnam Seminar, grading Fall 2020
- 21-325 Probability, grading Spring 2020
- 21-701 Discrete Mathematics (graduate), grading Fall 2019
- 21-325 Probability, grading Spring 2019
- 21-371 Algebraic Structures, grading Fall 2018
- 21-341 Linear Algebra, grading Fall 2018
- **Teaching assistant at PKU** 09/2017–01/2018
 - Calculus (college of engineering), grading and recitations Fall 2017

Publications

- **Sharp bounds on k -wise generalizations of oddtowns and eventowns**
 - With Minghui Ouyang and Lan Wei, [arXiv:2606.11139](#).
- **A dichotomy for hypergraph Zarankiewicz problems on axis-parallel boxes**
 - With Ting-Wei Chao, Hong Liu, Xichao Shu, and Shuaichao Wang, [arXiv:2604.20815](#).
- **Large grid subsets without many cospherical points**
 - With Zijian Xu, [arXiv:2506.18113](#).
- **Induced rational exponents and bipartite subgraphs in $K_{s,s}$ -free graphs**
 - With Jun Gao, Ruonan Li, and Hong Liu, [arXiv:2506.09020](#).
- **Set families: restricted distances via restricted intersections**
 - With Jun Gao, Hong Liu, Minghui Ouyang, and Qiang Zhou, [arXiv:2504.12296](#).
 - Appeared in [Math. Proc. Camb. Philos. Soc.](#), vol. 181(2), 2026.
- **Bipartite Turán problem on graph gluing**
 - With Jun Gao and Hong Liu, [arXiv:2501.12953](#).
 - Appeared in [Bull. Lond. Math. Soc.](#), vol. 57(12), 2025.
- **Many cliques with small degree powers**
 - With Ting-Wei Chao, Zijun Shen, and Ningyuan Yang, [arXiv:2410.04744](#).
- **Empty red-red-blue triangles**
 - With Ting-Wei Chao and Zhuo Wu, [arXiv:2409.17078](#).
 - The main result of the paper appeared in “A note on empty balanced tetrahedra in two-colored point sets in $\mathbb{R}^3$ ” by Díaz-Bañez, Fabila-Monroy, and Urrutia.

- **Saturation results around the Erdős–Szekeres problem**
 - With Gabór Damásdi, Manfred Scheucher, and Ji Zeng, [arXiv:2312.01223](#).
 - Preliminary version appeared in [SoCG 2024](#).
 - Appeared in [European J. Combin.](#), vol. 131, #104236, 2026.
- **Rainbow even cycles**
 - With Zijian Xu, [arXiv:2211.09530](#).
 - Appeared in [SIAM J. Discrete Math.](#), vol. 38(2), 2024.
- **Convex polytopes in restricted point sets in $\mathbb{R}^d$**
 - With Boris Bukh, [arXiv:2204.02487](#).
 - Appeared in [Adv. Comb.](#), 2025:1.
- **A simple proof of the Gan–Loh–Sudakov conjecture**
 - With Ting-Wei Chao, [arXiv:2201.05181](#).
 - Appeared in [Electron. J. Combin.](#), vol. 29(3), #3.59, 2022.
- **On the stability of graph independence number**
 - With Zhuo Wu, [arXiv:2102.13306](#).
 - Appeared in [SIAM J. Discrete Math.](#), vol. 36(1), 2022.
- **Longest common subsequences between words of very unequal length**
 - With Boris Bukh, [arXiv:2009.05869](#).

Presentations

- **k -wise odd-even towns**
 - Discrete Mathematics Seminar at IBS, Daejeon, Korea July 2026
- **Zarankiewicz for axis-parallel boxes**
 - 2026 CSIAM Annual Conference in Shijiazhuang, China May 2026
- **Zarankiewicz for axis-parallel boxes**
 - Combinatorics Seminar at SDU, Jinan, China May 2026
- **Set families: restricted distances via restricted intersections**
 - Atlanta Discrete Analysis Seminar at Emory, Atlanta, USA April 2026
- **Set families: restricted distances via restricted intersections**
 - Mingshu Seminar at TYUT, Jinzhong, China November 2025

- **Set families: restricted distances via restricted intersections**
 - Lectures of the New Stars at SDU, Jinan, China November 2025
- **Set families: restricted distances via restricted intersections**
 - Combinatorics and IT Seminar at Xidian (online), Xi'an, China November 2025
- **Large grid subsets without many cospherical points**
 - Seminar in Discrete Mathematics at Umeå University, Umeå, Sweden October 2025
- **Large grid subsets without many cospherical points**
 - CoGe Seminar at HUN-REN, Budapest, Hungary October 2025
- **Bipartite Turán problem on graph gluing**
 - School of Mathematics and Physics Seminar at XJTLU, Suzhou, China August 2025
- **Bipartite Turán problem on graph gluing**
 - School of Mathematical Sciences Seminar at PKU, Beijing, China June 2025
- **Set families: restricted distances via restricted intersections**
 - The 34th KIAS Combinatorics Workshop in Jeju, Korea May 2025
- **Maximizing the number of cliques in a graph of given degree sequence ℓ^p -norm**
 - Yongjiang Mathematics Forum at NBU, Ningbo, China November 2024
- **Lectures on Erdős–Szekeres combinatorial geometry results and related problems**
 - Combinatorics Seminar at NKU, Tianjin, China November 2024
- **Maximizing the number of cliques in a graph of given degree sequence ℓ^p -norm**
 - Mathematical Sciences Seminar at TYUT (online), Taiyuan, China November 2024
- **Maximizing the number of cliques in a graph of given degree sequence ℓ^p -norm**
 - Lectures of the New Stars at SDU, Jinan, China November 2024
- **Saturation around the Happy Ending**
 - Combinatorics Seminar at NPU, Xi'an, China November 2024
- **Maximizing the number of cliques in a graph of given degree sequence ℓ^p -norm**
 - Combinatorics Seminar at NPU, Xi'an, China November 2024
- **Saturation around the Happy Ending**
 - The Ninth Qilu Youth Forum at SDU (online), Jinan, China October 2024

- **Saturation around the Happy Ending**
 - SoCG 2024 in Athens, Greece June 2024
- **Saturation around the Happy Ending**
 - The 31st KIAS Combinatorics Workshop in Jeju, Korea May 2024
- **Convex polytopes in non-elongated point sets in $\mathbb{R}^d$**
 - Discrete Mathematics Seminar at IBS, Daejeon, Korea January 2024
- **Convex polytopes in non-elongated point sets in $\mathbb{R}^d$**
 - Erdős Center Seminar at the Rényi Institute, Budapest, Hungary October 2023
- **Rainbow even cycles**
 - AMS Spring Southeastern Sectional Meeting at GIT, Atlanta, USA March 2023
- **A simple proof of the Gan–Loh–Sudakov conjecture**
 - Graduate Students Seminar at CMU, Pittsburgh, USA February 2023
- **On the stability of graph independence number**
 - SIAM Conference (DM22) at CMU, Pittsburgh, USA June 2022
- **Convex polytopes in non-elongated point sets in $\mathbb{R}^d$**
 - ACO Seminar at CMU, Pittsburgh, USA April 2022
- **Convex independence and convex holes**
 - Graduate Students Seminar at CMU, Pittsburgh, USA October 2019
- **Random permutations and extremal set problems**
 - Graduate Students Seminar at CMU, Pittsburgh, USA February 2019
- **Domino Tilings of Aztec Diamonds**
 - Graduate Students Seminar at CMU, Pittsburgh, USA October 2018

Honors and awards

- Team Gold Award and Analysis Bronze Award, S.-T. Yau Maths Contest Beijing, 2017
- First Prize, Chinese Mathematics Competitions (CMC) Fuzhou, 2016
- Silver Medal, Chinese Mathematical Olympiad (CMO) Nanjing, 2013

Services

- Anonymous journal referee
 - *Discrete Comput. Geom.* (2023, 2024)
 - *J. Graph Theory* (2024)
 - *SIAM J. Discrete Math.* (2024)
 - *Amer. Math. Monthly* (2025)
 - *Adv. Comb.* (2025)
 - *Electron. J. Comb.* (2025)
 - *Israel J. Math.* (2026)
 - *J. Combin. Theory Ser. B* (2026)
- ECOPRO Student Research Program instructor Summer 2024 & Summer 2025
 - Initiated research publication [arXiv:2606.11139](#) with visiting students.
 - Initiated research publication [arXiv:2504.12296](#) with visiting students.
 - Initiated research publication [arXiv:2410.04744](#) with visiting students.
- Math Olympiad Summer Program (MOP) instructor Summer 2022
 - Trained the USA national team at International Math. Olympiad (IMO).
 - Gave lectures on algebra and number theory under the direction of Po-Shen Loh.
- Western PA ARML instructor 2019–2022
 - Trained the Western Pennsylvania Team at American Regions Math. League (ARML).
 - Gave lectures on combinatorics and number theory.